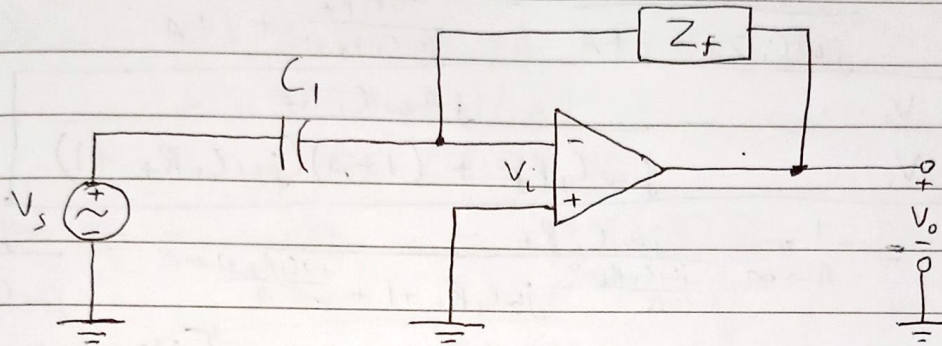


NESS M2023

HW4: CircuitsCHAPTER - 10

Q 59.



- a) A nodal equation at the negative op amp terminal V_n gives,
- $$(j\omega C_1)(V_n - V_s) + \frac{V_n - V_o}{Z_f} = 0$$

Since the positive op amp terminal is grounded, $V_p = 0$, then $V_i = V_n$ and $V_n = -V_o/A$

$$\Rightarrow (j\omega C_1) \left(-\frac{V_o}{A} - V_s \right) + \frac{-\frac{V_o}{A} - V_o}{Z_f} = 0$$

$$\Rightarrow \left(-\frac{j\omega C_1}{A} + \frac{-\frac{1}{A} - 1}{Z_f} \right) V_o - j\omega C_1 V_s = 0$$

$$\Rightarrow -\frac{j\omega C_1 Z_f + 1 + A}{AZ_f} V_o = j\omega C_1 V_s$$

$$\Rightarrow \frac{V_o}{V_s} = -\frac{jA\omega C_1 Z_f}{j\omega C_1 Z_f + 1 + A}$$

Now, $Z_f = R_f$ (given)

$$\Rightarrow \boxed{\frac{V_o}{V_s} = -\frac{jA\omega C_1 R_f}{j\omega C_1 R_f + 1 + A}}$$

$$\Rightarrow \lim_{A \rightarrow \infty} \frac{V_o}{V_s} = -\lim_{A \rightarrow \infty} \frac{j\omega C_1 R_f}{\frac{j\omega C_1 R_f + 1 + A}{A} + 1} = -j\omega C_1 R_f \quad [\text{VERIFIED}]$$

b) The impedance of a resistor R_f parallel to a capacitor C_f is given by,

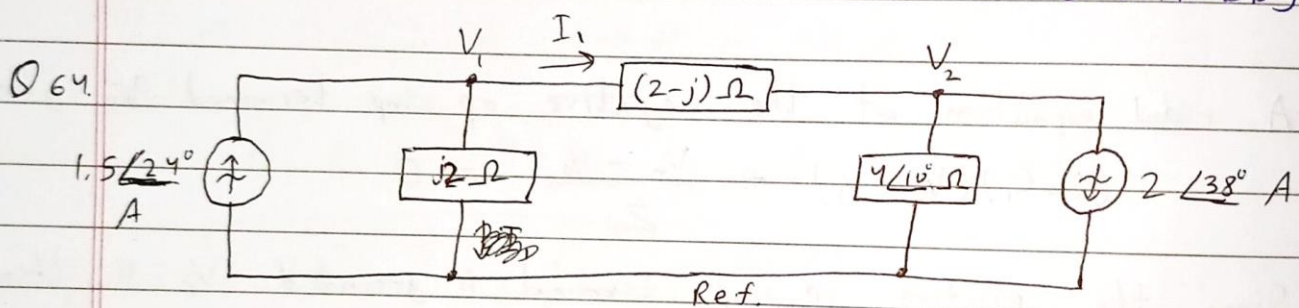
$$Z_f = \frac{R_f \times \frac{1}{j\omega C_f}}{R_f + \frac{1}{j\omega C_f}} = \frac{R_f}{\frac{j\omega C_f R_f + 1}{j\omega C_f R_f}} = \frac{j\omega C_f R_f}{j\omega C_f R_f + 1}$$

$$\therefore \frac{V_o}{V_s} = \frac{-jA\omega C_f Z_f}{j\omega C_f Z_f + 1 + A} = \frac{-jA\omega C_f R_f}{j\omega C_f R_f + 1 + A}$$

$$\Rightarrow \boxed{\frac{V_o}{V_s} = -\frac{jA\omega C_f R_f}{j\omega C_f R_f + (1+A)(j\omega C_f R_f + 1)}}$$

$$\Rightarrow \lim_{A \rightarrow \infty} \frac{V_o}{V_s} = -\lim_{A \rightarrow \infty} \frac{j\omega C_f R_f}{\frac{j\omega C_f R_f}{A} + j\omega C_f R_f + 1 + \frac{j\omega C_f R_f + 1}{A}} = \frac{-j\omega C_f R_f}{j\omega C_f R_f + 1}$$

[VERIFIED]



For Thévenin impedance, open the current sources.

$$\cancel{Z_{th}} \Rightarrow Z_{th} = j2 + 4\angle 10^\circ$$

$$\Rightarrow Z_{th} = j2 + 4\cos 10^\circ + j4\sin 10^\circ = 4.77\angle 34.32^\circ \Omega$$

$$\Rightarrow \boxed{Z_{th} = 4.77\angle 34.32^\circ \Omega}$$

$$\text{Now, } V_1 = (1.5\angle 24^\circ)(j2) = (1.5\angle 24^\circ)(2\angle 90^\circ) = 3\angle 114^\circ \text{ V}$$

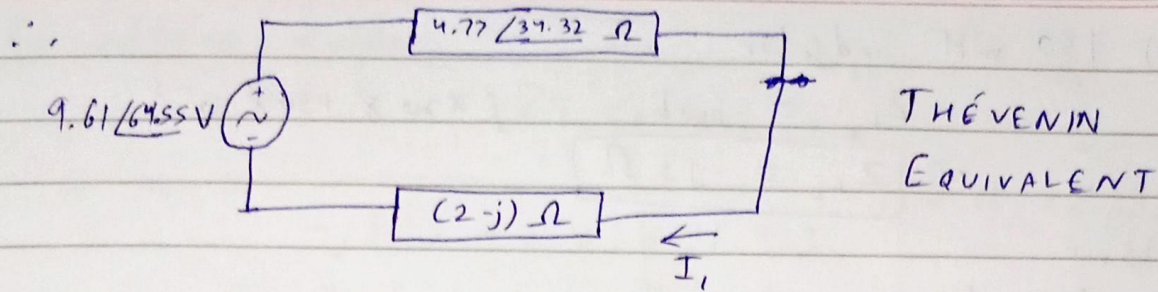
$$\text{and, } V_2 = -(2\angle 38^\circ)(4\angle 10^\circ) = -8\angle 48^\circ \text{ V}$$

[removing/isolating $(2-j)\Omega$ impedance to calculate Thévenin equivalent]

$$\text{Thévenin voltage, } V_{th} = V_1 - V_2 = [3\angle 114^\circ - (-8\angle 48^\circ)] \text{ V}$$

$$= 4.13 + j8.68$$

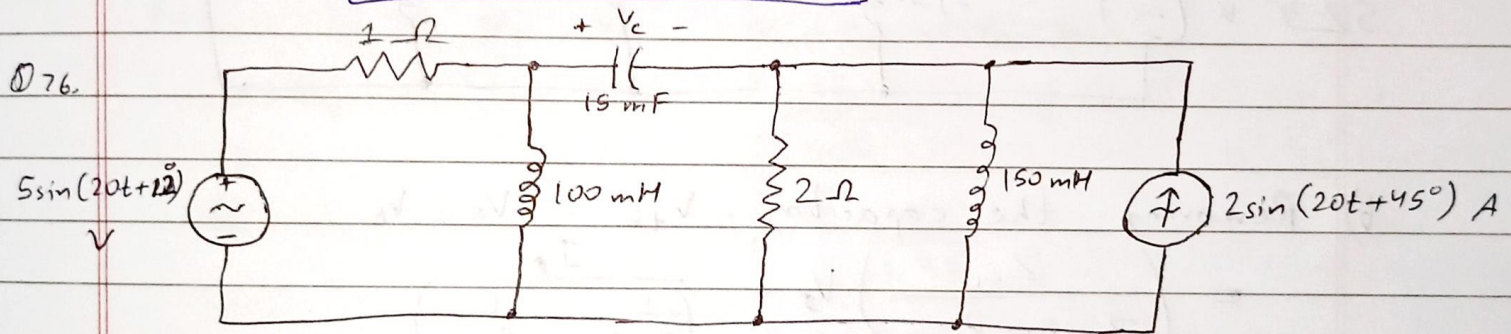
$$\Rightarrow \boxed{V_{th} = 9.61\angle 64.55^\circ \text{ V}}$$



$$\begin{aligned} Z_{total} &= 3.94 + j2.69 + 2 - j \\ &= 5.94 + j1.69 \\ &= 6.176 \angle 15.88^\circ \Omega \end{aligned}$$

$$\Rightarrow I_1 = \frac{V_{th}}{Z_{total}} = \frac{9.61 \angle 64.55^\circ}{6.176 \angle 15.88^\circ}$$

$$\Rightarrow \boxed{I_1 = 1.55 \angle 48.67^\circ \text{ A}}$$



a) i) Voltage source:

$$v_s = 5 \sin(20t + 12^\circ) \text{ V}$$

$$\Rightarrow V_s = 5 \angle (-90^\circ + 12^\circ) \text{ V}$$

$$\Rightarrow \boxed{V_s = 5 \angle -78^\circ \text{ V}}$$

ii) Current source:

$$i_s = 2 \sin(20t + 45^\circ) \text{ A}$$

$$\Rightarrow I_s = 2 \angle (-90^\circ + 45^\circ) \text{ A}$$

$$\Rightarrow \boxed{I_s = 2 \angle -45^\circ \text{ A}}$$

iii) 100 mH inductor:

$$Z_{L_1} = j\omega L_1 = j \times 20 \times 100 \times 10^{-3}$$

$$\Rightarrow \boxed{Z_{L_1} = j2 \Omega}$$

iv) 150 mH inductor :

$$Z_{L_2} = j\omega L_2 = j \times 20 \times 150 \times 10^{-3}$$

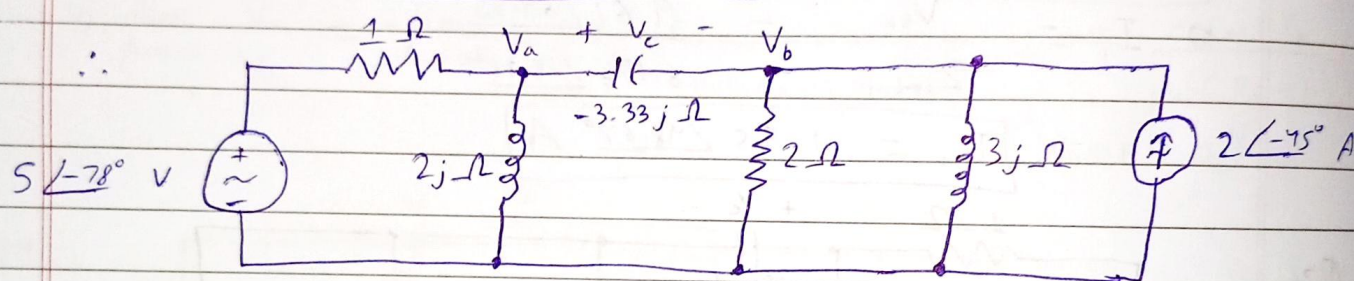
$$\Rightarrow \boxed{Z_{L_2} = j3 \Omega}$$

15 mF

v) Capacitor :

$$Z_c = \frac{1}{j\omega C} = \frac{1}{j \times 20 \times 0.015}$$

$$\Rightarrow \boxed{Z_c = -j3.33 \Omega}$$



b) Removing the capacitor, $V_{th} = V_a - V_b$

$$= \left(\frac{Z_{L_1}}{Z_{L_1} + R_{1\Omega}} \right) V_s - \frac{I_s}{\left(\frac{1}{R_{2\Omega}} + \frac{1}{Z_{L_2}} \right)}$$

$$= \frac{j2}{j2 + 1} (5 \angle -78^\circ) - \frac{(2 \angle -45^\circ)}{\left(\frac{1}{2} + \frac{1}{j3} \right)}$$

$$\Rightarrow \boxed{V_{th} = 2.8835 \angle -99.494^\circ \text{ V}}$$

For Thévenin impedance, short V_s and open I_s .

$$\Rightarrow Z_{th} = \left(\frac{1}{R_{1\Omega}} + \frac{1}{Z_{L_1}} \right)^{-1} + \left(\frac{1}{R_{2\Omega}} + \frac{1}{Z_{L_2}} \right)^{-1}$$

$$= \left(1 + \frac{1}{j2} \right)^{-1} + \left(\frac{1}{2} + \frac{1}{j3} \right)^{-1}$$

$$= \frac{j2}{j2 + 1} + \frac{j6}{j3 + 2}$$

$$\Rightarrow \boxed{Z_{th} = 2.185 + j1.323 \Omega}$$

By voltage divider, $\frac{Z_c}{Z_c + Z_{th}} V_{th} = V_c$

$$\Rightarrow V_c = \frac{-j3.33}{-j3.33 + 2.185 + j1.323} (2.8835 \angle -99.444^\circ)$$

$$= 3.236 \angle -146.925^\circ \text{ V}$$

$$\therefore \boxed{V_c = 3.236 \sin(20t - 146.925^\circ) \text{ V}}$$

c) KCL at V_a :

$$\frac{V_a - V_s}{1 \Omega} + \frac{V_a}{2j \Omega} + \frac{V_a - V_b}{-3.33j \Omega} = 0$$

But, $V_c = V_a - V_b \Rightarrow V_b = V_a - V_c$

$$\Rightarrow V_a - V_s - \frac{jV_a}{2} + \frac{jV_c}{3.33} = 0$$

$$\Rightarrow V_a \left(\frac{2-j}{2} \right) = V_s - \frac{jV_c}{3.33}$$

$$= 5 \angle -78^\circ - \frac{j 3.236 \angle -146.925^\circ}{3.33}$$

$$= 0.5092 - j4.07644 = V_a(1 - j0.5)$$

$$\Rightarrow V_a = 2.038 + j3.0575 \text{ V}$$

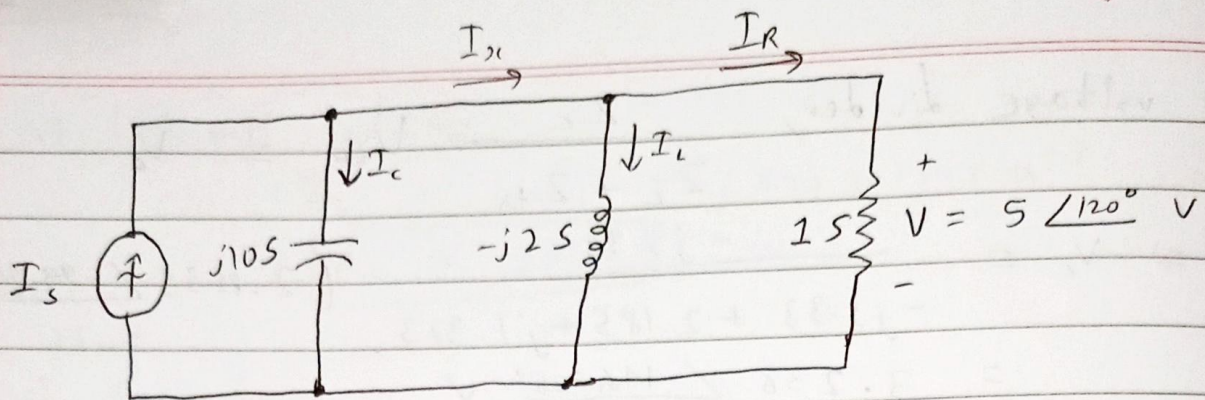
$\therefore$ Current through the terminal of V_s ,

$$i_+ = \frac{V_a - V_s}{R_{1\Omega}} = 5 \angle -78^\circ - (2.038 + j3.0575)$$

$$= -0.998 - j1.833 \text{ A}$$

$$\Rightarrow \boxed{I_+ = 2.0875 \angle -118.573^\circ \text{ A}}$$

Q71.

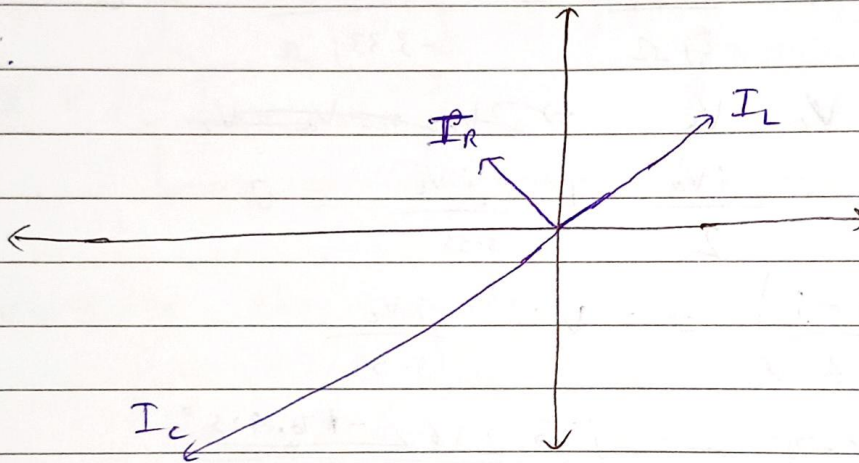


a)

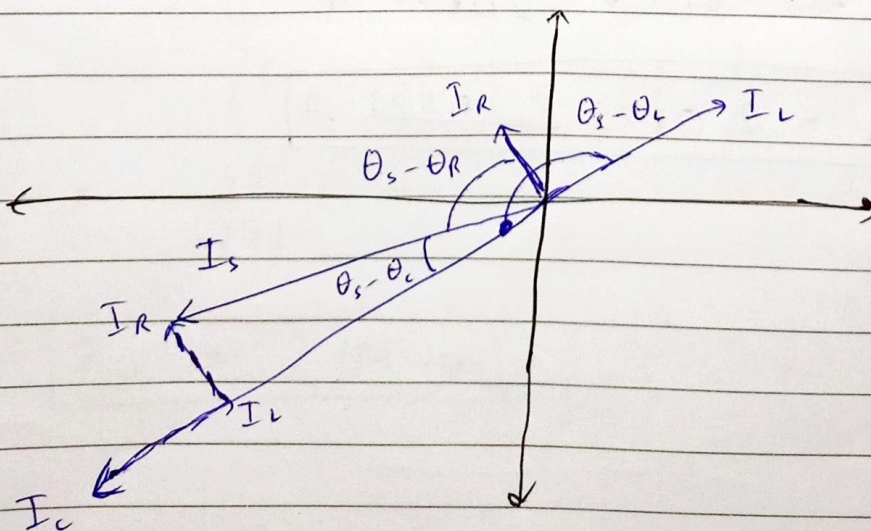
$$I_R = 1 \cdot 5 \angle 120^\circ = 5 \angle 120^\circ \text{ A}$$

$$I_L = (-j2)(5 \angle 120^\circ) = 10 \angle 30^\circ \text{ A}$$

$$I_C = (j10)(5 \angle 120^\circ) = 50 \angle -150^\circ \text{ A}$$



- b) By KCL, $I_s = I_R + I_C = I_R + I_L + I_C$
 Since I_L & I_C are anti-parallel, the tail of I_C can be coincided with the head of I_L , via shifting I_L , so that $I_L + I_C$ has its tail at origin and head at I_L 's head. To add I_R , shift it so its tail is at I_L 's head.



$$\Rightarrow \theta_{S/C} = \theta_s - \theta_c = -7.1^\circ$$

$$\theta_{S/R} = \theta_s - \theta_R = 82.9^\circ$$

$$\theta_{S/L} = \theta_s - \theta_L = 172.9^\circ$$

CHAPTER - II

Q51. $V_{rms} = 1200 \text{ V}$, $I_{rms} = 10 \text{ A}$, $f = 50 \text{ Hz}$, $PF = 0.9$

a) $V_{peak} = \sqrt{2} V_{rms} = \sqrt{2} (1200) \Rightarrow V_{peak} = 1697.06 \text{ V}$

b) $I_{peak} = \sqrt{2} I_{rms} = \sqrt{2} (10) = 14.14 \text{ A}$

$$PF = \cos(\theta - \phi) \Rightarrow |\theta - \phi| = \cos^{-1} PF$$

Since PF is lagging, $\phi < \theta$. Also $\theta = 0$ (voltage source)

$$\Rightarrow \phi < 0 \Rightarrow \phi = -\cos^{-1}(0.9) \Rightarrow \phi = -25.8^\circ$$

$$\therefore v(t) = V_{peak} \cos(2\pi f t) \text{ V}$$

$$= 1697.07 \cos(100\pi t) \text{ V}$$

$$i(t) = I_{peak} \cos(2\pi f t + \phi) \text{ A}$$

$$= 14.14 \cos(100\pi t - 25.8^\circ) \text{ A}$$

$$\Rightarrow p(t) = v(t) \cdot i(t) = 23.997 \cos(100\pi t) \cos(100\pi t - 25.8^\circ) \text{ kW}$$

$$\therefore p(t = I_{rms}) = 22.61 \text{ kW}$$

c) Apparent power, $|S| = V_{rms} \cdot I_{rms} = 1200 \times 10 \text{ VA}$
 $\Rightarrow |S| = 12 \text{ kVA}$

d) Reactive power, $Q = |S| \sin \theta_s = |S| \sin(\theta - \phi)$
 $= 12 \times 10^3 \times \sin(25.8^\circ)$

$$\Rightarrow Q = 5.223 \text{ kVAR}$$

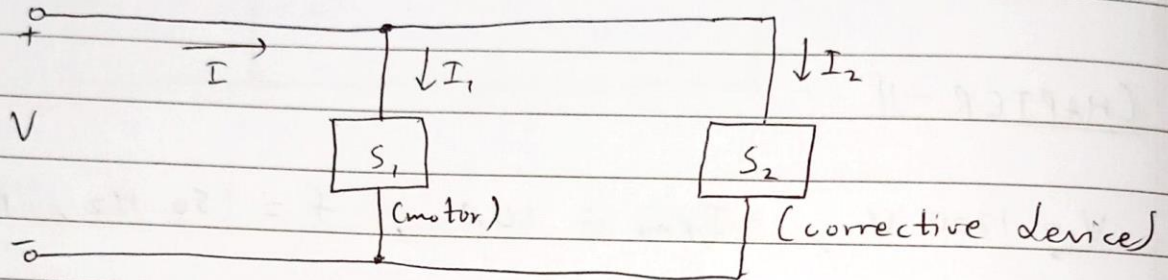
e) Load impedance, $Z_L = \frac{V_{peak} \angle \theta}{I_{peak} \angle \phi} = \frac{1697.06 \angle 0^\circ}{14.14 \angle -25.8^\circ}$

$$\Rightarrow Z_L = 120.02 \angle 25.8^\circ \Omega$$

$$f) S = |S| \angle \theta_s = |S| \angle \theta - \phi = 12 \times 10^3 \angle 25.8^\circ \text{ VA}$$

$$\Rightarrow \boxed{S = 12 \angle 25.8^\circ \text{ kVA}}$$

Q43.



$$S_{\text{motor}} = 150 \angle 24^\circ \text{ VA} = S$$

$$a) PF = \cos \theta_s = \cos 24^\circ \quad [\text{without connecting } S_2]$$

$$\Rightarrow \boxed{PF = 0.914 \text{ lagging}}$$

$$b) \text{ Real power drawn, } P = \text{Re}\{S\} = |S| \cos(\theta_s)$$

$$\Rightarrow P = 150 \cos(24^\circ) \Rightarrow P = 137.03 \text{ W}$$

$$\text{New power factor, } PF' = \cos(\theta_s')$$

$[\theta_s' \rightarrow \text{phase after connecting } S_2]$

$$\Rightarrow \theta_s' = \cos^{-1}(PF') = \cos^{-1}(0.98)$$

$$\Rightarrow \theta_s' = 11.5^\circ$$

The real power does not change.

$$\therefore P = |S'| \cos(\theta_s')$$

$$\Rightarrow |S'| = \frac{P}{\cos \theta_s'} = \frac{137.03}{0.98} = 139.83 \text{ VA}$$

$$\therefore S' = |S'| \angle \theta_s' = 139.83 \angle 11.5^\circ \text{ VA}$$

$$\text{Now, } S' = S_1 + S_2 \Rightarrow S_2 = S' - S_1 = S' - S$$

$$\Rightarrow S_2 = 139.83 \angle 11.5^\circ - 150 \angle 24^\circ = -j33.13 \text{ VA}$$

$$\text{Also, } S_2 = VI_2^*$$

$$\Rightarrow \cancel{Z_2} = \frac{V}{I_2} \Rightarrow I_2 = \left(\frac{S_2}{V} \right)^*$$

$$I_2 = \left(\frac{-j33.13}{230} \right)^* \Rightarrow I_2 = j0.144 \text{ A rms}$$

Using $V = 230 \text{ V rms}$,

$$Z_2 = \frac{V}{I_2} = \frac{230}{j0.144}$$

$$\Rightarrow \boxed{Z_2 = -j1597 \Omega}$$

- c) To obtain a leading PF for the source, its reactive power component should be negative. It is possible to adjust Z_2 so that its reactive power is more negative than the motor reactive power is positive. Thus, the input impedance, ~~with~~ having a capacitive component, has negative reactive power.

physically

$\therefore$ Yes, it is possible to obtain a leading PF for the source, by adjusting Z_2 .